

CONTINUATION — HelixTranslate

session-resumption file

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Session 2026-06-14v — cross-submodule bug-hunt campaign (§11.4.28 equal-codebase)

Parallel subagent-driven (§11.4.70/§11.4.103) hunt across owned submodules — each bug reproduce-first RED (§11.4.115) → fix → GREEN → mutation-proven (revert→FAIL/race→restore→PASS), verified by the conductor against repo state before commit (§11.4.142), committed in each submodule's own repo (raw git, FF-only, no force §11.4.113, multi-mirror §2.1). **18 genuine mutation-proven bugs fixed + pushed; vision_engine's was independently fixed upstream by a parallel session (64b7d08) so it was integrated, NOT double-fixed.** Host API rate-limit (prior session's 6-subagent killer) had recovered — proven by a probe wave that all survived.

Submodule	bug(s) fixed	commit (after FF)
llm_provider	health monitor recovered on cumulative not consecutive successes	f58559f
llm_provider	claude stream fake-success-on-error; openrouter 4096B SSE truncation; zen dropped system-prompt; zen sessionID data race	325a469
doc_processor	truncate byte-slice corrupts multibyte (Cyrillic) Feature.Description	eec2d3e
doc_processor	i18n interpolate re-substitutes placeholders, order-dependent Cyrillic corruption	fc4fed8
llm_orchestrator	SimpleAgentPool.Release returns untracked agents (double-handout + cross-pool leak)	1fac8b1
llm_orchestrator	RoundRobinSelector never selects a lazy pool with spare build capacity (dead lazy-build path)	4e08b93
llms_verifier	HTTP/3-disabled nil-panic + ignored maxScore ceiling	6e5ab147
llms_verifier	nil circuit-breaker deref panic ×3 sites + concurrent map write under RLock (data race)	6f19503b
docs_chain	corrupt SQLite node silently read as empty DB (SSoT-destroying)	47275bc
docs_chain	verify masked multi-level staleness — CI gate passed a stale artifact	ad8b9ff
challenges		e4ae4ef

Submodule	bug(s) fixed	commit (after FF)
	go/cargo test-JSON parsers silently drop failures after >64KB line (PASS-bluff in the anti-bluff harness)	
containers	boot summary counter corruption (Started=-1) + health check ignored ctx cancellation	f3bfb2
MAIN pkg/ distributed	slice-header data race (remoteInstances) + BatchProcessor processFn-under-lock deadlock + version_manager backups fatal concurrent-map-write + alerts slice race — all —race-proven	3a5b435
MAIN internal/ verifier	GetPreferences dropped ALL models without a prior RefreshScores (used uncached score vs threshold); empty verified-models result never cached (TTL defeated)	(wave 7)
MAIN pkg/ ebook + pkg/fb2	TXT >64KiB single-line total content loss; EPUB multiline <head> leaks <title> text into chapter; FB2 stanza title/subtitle dropped; FB2 cite/epigraph nested <poem> dropped — round-trip data-loss class	(wave 7)
containers pkg/ crossbuild	copyFile false-FAIL on directory artifacts (jpackage app-images) — successful build reported as failure (§11.4.1), all 3 non-Apple backends	ebf2641
MAIN pkg/ storage	cache-key NUL-delimiter collision (distinct (lang,provider,model,text) tuples → same sha256 → WRONG cached translation served) across SQLite/Postgres/Redis; length-prefix fix + §11.4.120 reconcile	(wave 8)

Inline non-finding (§11.4.6): pkg/script (Serbian Cyrillic[?]Latin) audited — conversion correct for its rule-based scope (all 30 letters both directions, uppercase-digraph all-caps logic sound, rune-iterated); the loanword nj/dž disambiguation (e.g. *injekcija*→*инјекција*) is an inherent rule-based-transliteration limitation needing an exception dictionary, NOT a narrow bug — no fix manufactured.

Main-repo + submodule-pointer sync (operator directive 2026-06-14): main repo committed the 4 pkg/distributed concurrency fixes (above) + advanced the 8 submodule gitlink pointers (challenges/containers/doc_processor/docs_chain/llm_orchestrator/llm_provider/llms_verifier/vision_engine) to their pushed HEADs so parent [?] submodules are in sync. NOT staged: helix_qa (m, another session's in-flight work, §11.4.119); tracked rebuilt binaries preparation-translator/translate-ssh/unified-translator + untracked hash/workable-items (build artifacts, §11.4.30 — never committed).

Anti-bluff non-findings (honest, §11.4.6 — did NOT manufacture fixes): llm_provider generic.go empty-choices→empty-content is by-design tested contract; numerous adapters/packages audited clean with enumerated coverage; vision_engine gocv-build bug already fixed upstream.

§11.4.147 NOTE (operator awareness): a SEPARATE sibling clone exists at /Volumes/T7/Projects/llm_orchestrator (outside the parent tree). A subagent accidentally wrote the first RoundRobinSelector fix THERE; I left it untouched (unknown provenance) and re-applied the fix correctly inside the submodule (committed 4e08b93). That sibling now holds a stale, superseded, uncommitted change — safe to discard, but flagged so it isn't mistaken for live work.

Session 2026-06-14u — wave 14: governance depth + coverage + e2e build-fix (HEAD 84f7a46)

- **22a32e3** §11.4.93 SQLite SSoT extended: sync db-to-md (byte-stable summary round-trip), item_history table + record-event, diff. (Full per-item-prose body round-trip honestly noted as not-yet-in-schema.)
- **c049907** pre-build gates +2 → **8-gate suite, all green** (CM-DOC-SIBLING-SYNC §11.4.65 + CM-NO-FORCE-PUSH-ABSOLUTE §11.4.113), all 8 paired mutations pass; caught a real gate-regex bug pre-commit.
- **fef82cc** real e2e pipeline coverage: test/integration/pipeline_roundtrip_test.go (FB2/TXT × EPUB/FB2 + httpstest-provider HTTP path), asserts actual output-artifact content, empty-stub mutation-proven.
- **84f7a46** fixed a pre-existing §11.4.1 build break: test/e2e package didn't compile (unused discovery import) — removed; go vet -tags e2e ./test/e2e/ now clean.

FINAL sweep GREEN at HEAD 84f7a46 (quiescent): go test ./... -p 1 = **58 ok / 0 FAIL**; 8-gate pre-build suite PASS; build+vet clean; all FF-pushed both upstreams (no force §11.4.113).

NEW tracked gap (for follow-on): scripts/commit_all.sh does NOT auto-invoke sync_all_markdown_exports.sh, so doc commits leave stale .html/.pdf siblings (CM-DOC-SIBLING-SYNC catches them post-hoc; the gate agent had to regen CONTINUATION siblings manually). Follow-on: wire the export-sync into commit_all.sh (§11.4.75 layer-2). Until then, regen siblings when committing a doc.

SESSION FINAL STATE (unchanged from Rev 43 + wave 14) — autonomous queue DRAINED

Loop correctly rests on operator decisions. Remaining items (all operator-gated or larger multi-session follow-ons) tracked in docs/WORKING_PLAN.md + docs/Issues.md ATM-065..081: version number · pkg/hash remove-vs-relocate · provider keys/balance → allowlist audits · design calls · fuller §11.4.93 body round-trip · commit_all export-sync wiring · owned-submodule waves (another session active on helix_qa) · full §11.4.27 test-type matrix · full §11.4.40 retest → §11.4.151 release tag. Resume: point a fresh session at this file + docs/WORKING_PLAN.md.

Session 2026-06-14t — plan-execution wave 13; autonomous queue DRAINED (HEAD 6444e38)

Wave 13 closed the last clean autonomous WORKING_PLAN items (3 background subagents, disjoint, all real + proven):

- **e623ffb** §11.4.93 workable-items SQLite SSoT: cmd/workable-items Go tool (sync md-to-db/validate/list) + tracked docs/workable_items.db

(§11.4.95). Real run: 81 items (64 fixed + 17 open), validate exit 0, verified via sqlite3 SELECT; mutation-proven. (Fuller schema — item_history/obsolete_details/db-to-md/fingerprint — is a tracked follow-on.)

- **78023a7** P4.2 +2 gates: CM-TRACKER-DOCS-PRESENT + CM-ATM-TICKET-IDS → **pre-build suite now 6 gates, all green**, each paired-mutation-proven.
- **6444e38** P4.3/ATM-076 §11.4.65 doc-export: scripts/testing/sync_all_markdown_exports.sh (pandoc html + weasyprint pdf, idempotent, --check gate mode, §11.4.106 honest tool-absence) + **225 siblings generated (0 failures, 0 stale)** for 120 in-scope docs. Caught + excluded a stray docs/research/helix_qa/ workspace dump (§11.4.30 guard).

FINAL integration sweep GREEN at HEAD 6444e38 (quiescent): go
test ./... -p 1 = **58 ok / 0 FAIL**; 6-gate pre-build suite PASS; build+vet clean.
All FF-pushed both upstreams (no force §11.4.113).

SESSION FINAL STATE — autonomous queue DRAINED

Delivered: ~30 genuine mutation-proven bug fixes (13 waves; main-module bug-hunt SATURATED per §11.4.118 completeness audit); format matrix complete (6 in × 5 out); PDF+DOCX input revived; deepseek-v4 allowlist current; version single-source (5 binaries); workable-items tracker constellation (81 ATM tickets) + SQLite SSoT; 6-gate mutation-proven pre-build suite; §11.4.65 doc-export (225 siblings); perf/stress/chaos coverage (5 hot-path packages); docs/WORKING_PLAN.md (canonical unfinished+known-issues inventory).

REMAINING — all operator-gated or larger tracked follow-ons (NOT autonomously closable; see docs/WORKING_PLAN.md + docs/Issues.md ATM-065..081):

1. **OPERATOR DECISIONS:** P1.0 version number (2.3.0 vs 3.0.0; wiring done) · P3.1 pkg/hash remove-vs-relocate · P1.1 provider keys/balance (OPENAI/ANTHROPIC absent, GEMINI invalid, ZHIPU no-balance) → unblocks stale-allowlist audits (deepseek pattern proven) · P2.x design calls (inert flags, MinScoreThreshold scale, reasoning-model structured-content, markdown first-class input).
2. **LARGER TRACKED FOLLOW-ONS:** fuller §11.4.93 schema (item_history/db-to-md/fingerprint) · owned-submodule bug-hunt waves (§11.4.28 — ANOTHER SESSION ACTIVE on helix_qa, coordinate per §11.4.119) · full §11.4.27 test-type matrix + HelixQA/Challenges banks · P2.6 cmd/translator SSH download-dir (needs live SSH via §11.4.76 Containers).
3. **RELEASE:** P0.2 full §11.4.40 7-step retest → P0.3 §11.4.151 helix_translate-<version> tag (after the operator decisions + the test-type matrix).

Loop status (§11.4.94(A)/§11.4.101/§11.4.126): correctly rests on operator decisions — every remaining item is externally/operator-gated and explicitly tracked; nothing silently dropped. Resume by pointing a fresh session at this file + docs/WORKING_PLAN.md.

Session 2026-06-14s — WORKING_PLAN created + plan-execution begins (HEAD 06a8a16)

Operator mandate: comprehensive subagent-driven plan of all unfinished items + known issues, then execute. Bug-hunt declared SATURATED by the §11.4.118 completeness-critic (wave 10).

- **docs/WORKING_PLAN.md** (78ce2d0) — the canonical no-bluff inventory: P0 release blockers, P1 operator/credential-gated, P2 design decisions, P3 dead code, P4 governance gaps, P5 owned submodules, P6 full test-type coverage. Each item: WHAT/WHY-OPEN/EVIDENCE/SUBAGENT-TASK/ACCEPTANCE.
- **Wave 10 (2 fixes + saturation verdict):** e520c4c enhanceTranslation byte→rune capitalization (silently no-op for Cyrillic/accented first letters) + isRussianToSerbian empty-source (told LLM wrong source lang); 5b0dd17 the orphaned guard. docx_parser: no bug (exhaustive). Completeness-critic: main-module product bug-hunt SATURATED.
- **Wave 11 (3 plan items):** a36030e P0.1 version single-source (pkg/version/app.go AppVersion=2.3.0; 5 divergent binaries reconciled, runtime-verified, mutation-proven); 981ced9 P3.1 §11.4.124 pkg/hash investigation (dead package main duplicate of pkg/version.CodebaseHasher — finding doc written, NOT removed per §11.4.122, **operator decision queued: remove-as-duplicate vs relocate**); 279ac7c P4.2 first real pre-build gate suite (CM-GITIGNORE-PRECOMMIT-AUDIT + CM-NO-FAKES-BEYOND-UNIT + paired mutation meta-tests). 59819fa→06a8a16 §11.4.120 reconcile of stale version assertions in cmd/cli (v2.0.0→dynamic) + cmd/server (fixed a tautological Contains("v1.0.0", "1.0.0") §11.4.1 bluff). [Note: 59819fa committed a test before go test verified it — go build ./... skips _test.go; repaired fix-forward in 06a8a16. Lesson: use go vet/go test as the pre-commit compile check for test changes.]
- **§11.4.65 finding (P4.3):** only 3 of 87 tracked docs/*.md have .html/.pdf siblings — large export gap (needs pandoc/weasyprint or honest §11.4.106 ToolAbsent skip).
- **Wave 12 in flight:** P4.1 tracker constellation scaffold (Issues/Fixed/summaries + ATM-NNN) · P6 perf/benchmark + stress/chaos coverage · P4.2 gate-suite expansion (CM-SCRIPT-TARGET-SHELL-PARSEABLE + CM-VERSION-SINGLE-SOURCE).

Integration sweep GREEN at HEAD 06a8a16 (quiescent): go test ./... -p 1 = **57 ok / 0 FAIL**. build+vet clean. CAMPAIGN TOTAL: ~30 genuine mutation-proven fixes; main-module bug-hunt saturated; plan execution underway.

OPERATOR DECISIONS QUEUED: (1) P1.0 authoritative version number (2.3.0 vs 3.0.0); (2) P3.1 pkg/hash remove-vs-relocate; (3) P1.1 provider keys/balance (OPENAI/ANTHROPIC absent, GEMINI invalid, ZHIPU no balance) to unblock allowlist audits; (4) P2.x design decisions (inert flags, MinScoreThreshold scale, reasoning-content support). See docs/WORKING_PLAN.md.

Session 2026-06-14r — bug-hunt waves 8-9: 7 more genuine mutation-proven fixes (HEAD daa2f70)

Continued the §11.4.70/§11.4.103 parallel campaign (background subagents on disjoint cmd/ + pkg/ scopes; reproduce-first + mutation-proven + single-package -p 1/-race; FF-pushed, no force).

- **86d10d4** cmd/translate-ssh: -output ignored — translated EPUB delivered to the INPUT dir instead of the requested path, then printFinalReport claimed success pointing at a nonexistent file (wrong-delivery + false-success).
- **309ca91** cmd/cli: gemini missing from getAPIKeyFromEnv env map → GEMINI_API_KEY silently ignored (translation failed with a key present).
- **21c2460** cmd/grpc-server: documented env-var overrides (GRPC_ADDRESS/PORT, LOG_LEVEL, ENABLE_METRICS/REFLECTION) never implemented → silently ignored.
- **cde2e1a** cmd/deployment: handleStatus sliced ContainerID[:12] unconditionally → panic crashing the whole status command on any <12-char/empty ID.
- **93e89d7** cmd/grpc-server + cmd/monitor-server: wired the dead -max-connections flag (was hardcoded 50) + stopped swallowing monitor-server's router.Run() bind error (silent failed start).
- **daa2f70** logger: JSON reserved-key collision — a user field named level/message/timestamp overwrote the authoritative metadata → real severity/message/timestamp silently dropped.

Integration sweep GREEN at HEAD daa2f70 (quiescent): go test ./... -p 1 = **57 ok / 0 FAIL** (up from 55 — the wave fixes added committed tests to 5 cmd/packages that previously had none). build+vet exit 0.

CAMPAIGN TOTAL (waves 1-9): 25 genuine mutation-proven fixes (27 subagents). SESSION GRAND TOTAL ~32 real fixes. Wave 10 (completeness-critic audit + docx_parser + llm prompt/cache helpers) in flight to confirm saturation.

REMAINING (operator/design-gated, unchanged): OPENAI/ANTHROPIC keys absent; GEMINI_API_KEY invalid; ZHIPU out of balance (allowlist stale, can't verify); ~30 other provider allowlists need funded keys; verifier MinScoreThreshold 0-100-vs-0-10 scale (no caller, operator decision); -chunk-size inert flag (design); markdown first-class input (design); reasoning-model structured-content client support (design); cmd/translator intermediate-md download-dir inconsistency (needs live SSH); submodules (challenges/containers/helix_qa) being worked by another session — do not collide (§11.4.119). NO release tag yet (needs §11.4.40 full retest + operator confirm + §11.4.151 prefix).

Session 2026-06-14q — parallel subagent-driven bug-hunt campaign (§11.4.70/§11.4.103): 18 genuine mutation-proven fixes across 7 waves (HEAD 4bbac9a)

Operator directive: continuous endless loop with 3–4 parallel subagents on all parallelizable workable items, rock-solid evidence, zero bluff. Ran 7 waves (21 subagents on DISJOINT package scopes per §11.4.58/§11.4.119); every fix reproduce-first (§11.4.115) + mutation-proven (§1.1) + single-package -p

1(+race) validated; all FF-pushed to both upstreams (no force §11.4.113). 4 subagents honestly reported NO bug (events/websocket, storage, api/services) — no manufactured changes (§11.4.6).

The 18 fixes:

- **80d627b** hardware: `parseLscpuCores` undercounts cores on multi-socket hosts (ignored `Socket(s)`; dual-socket 16-core → 8).
- **0df25d9** preparation: coordinator never stamped `ChapterNum` → 0 → positional-fallback chapter-context mis-attribution after a failed chapter compacts.
- **c3117f8** batch: same-stem/different-ext inputs (`book.fb2+book.epub`) collide onto one output → data loss; reconciled the 3 stem-only tests (§11.4.120).
- **d0fe40a** security: `RefreshToken` minted fresh tokens from EXPIRED/no-expiry claims (auth-bypass, session resurrection); zero residue confirmed.
- **d8142e5** verifier: `/api/v1/verified-models` hard-errored on zero verified models (`{"models":null}` fell through to array-decode).
- **20beda7** distributed: `getFailureRate` fabricated 100% failure rate on zero/expired window → falsely traps coordinator in degraded mode.
- **8f52370** fb2: `<v>/<subtitle>/<text-author>` dropped inline-element text (verse/quote/attribution data loss).
- **a728e57** llm/ollama: dropped `temperature/max_tokens` (must be in options per official API §11.4.99) → non-deterministic output.
- **be81550** markdown: chapter XHTML `<title>` unescaped → malformed EPUB for titles with `&/</>`.
- **7434ee4** verification: `polishChapter/polishSectionRecursive` index-out-of-range panic when translated has fewer sections.
- **fd07e18** grpc: `SubscribeEvents` delivered lifecycle events with empty `session_id`.
- **107e570** ebook/epub_parser: chapter text shipped HTML entities literally (Tom & Jerry, `café`).
- **fa35d2e** config: nil-map panic in `LoadConfig` (omitted providers + any API-key env var set).
- **8316e3f** translator core: data race / concurrent-map-write crash in `BaseTranslator` cache+stats (race-proven).
- **c416bd8** ebook/epub_writer: cover image mislabeled `image/jpeg` for PNG/GIF/WEBP/SVG covers → broken cover.
- **36f740a** cmd/unified-translator: SSH provider used a hardcoded "ru→sr-cyrillic" prompt + hardcoded binary/model paths, ignoring all user flags (also carried the cmd/api-server health fix below, swept in by the wrapper).
- **(36f740a)** cmd/api-server: `healthCheck` returned 200 "healthy" regardless of gRPC backend state → derives from `conn.GetState()` → 503 when not Ready.
- **85b3362** cmd/markdown-translator: `-format` flag never validated → unsupported value prints success but writes no file (false-success bluff).
- **4bbac9a** .gitignore (§11.4.30): bare `api-server` pattern ignored the whole `cmd/api-server/` SOURCE dir (no repo-root binary exists; build output is `build/api-server`) → hid the package + blocked its tests (api-server had ZERO committed tests). Anchored to `/api-server`; committed the health regression guard.

Operator-flag (no production caller, design decision): verifier

MinScoreThreshold scale inconsistency — handler compares raw 0-100, adapter compares normalized 0-10; GetPreferences/GetProviderScore have no production caller; needs an operator decision on the canonical scale before either is wired (not guessed, §11.4.6/§11.4.120).

Process lesson (§11.4.119): the full `go test ./... -p 1` integration sweep includes pkg/version's CodebaseHasher which hashes the whole tree (.go + .md + docs/) — it MUST run on a QUIESCENT tree; running it concurrently with file-writing subagents produces a spurious pkg/version hash-mismatch FAIL (confirmed green isolated). Sweeps + CONTINUATION syncs are therefore serialized BETWEEN parallel waves.

Integration sweeps GREEN at each quiescent checkpoint: waves 1-5 → 54 ok/0; waves 6+7 → **55 ok/0 FAIL** (HEAD 4bbac9a; cmd/api-server now has committed tests). build+vet exit 0. **SESSION GRAND TOTAL: ~25 real fixes** (this campaign's 18 + earlier: PDF revival, format matrix + .html, deepseek-v4 allowlist, websocket race, ssh-test, 2 doc-syncs).

Session 2026-06-14p — Zhipu allowlist §11.4.150 finding (operator-blocked); autonomously-provable queue drained

- **Zhipu allowlist is stale (Operator-blocked, §11.4.21):** live Zhipu /models (account-authoritative, verified 2026-06-14) returns **glm-4.5, glm-4.5-air, glm-4.6, glm-4.7, glm-5, glm-5-turbo, glm-5.1** — NONE of our allowlisted glm-4 family (glm-4/glm-3-turbo/glm-4-plus/glm-4-flash/glm-4-air/glm-4-airx/glm-4-long/glm-4-flashx) are listed. The generic factory (llm.go:218) HARD-REJECTS unlisted models, so a funded account requesting a current model is rejected (this is what the earlier "zhipu rejects documented models" flag actually was). **Why not fixed autonomously:** the account is OUT OF BALANCE (chat API error 1113 "余额不足/insufficient balance") → I cannot obtain §11.4.123 translation proof NOR verify response-shape compatibility (the magistral lesson: glm-5 reasoning models may return structured-list content our string-content client cannot parse — unverifiable while balance-blocked). Blindly adding glm-5 could enable a client-incompatible model. **Operator-Block-Details:** WHAT — recharge the Zhipu account OR provide a funded ZHIPU_API_KEY; WHY — every self-resolution exhausted (key present + API reachable, but balance blocks all verification; cannot manufacture credit); UNBLOCK CONDITION — a funded zhipu key so the deepseek-pattern fix (verify-translate + string-content shape, then additive allowlist update + RED-proven gate guard) can be applied; WHO — operator.
- Contrast: **deepseek** had a funded key → fully provable → FIXED this session (0fd1a34). The verify-then-fix pattern is established; it just needs a funded provider account.

AUTONOMOUSLY-PROVABLE QUEUE DRAINED. All clearly-actionable, evidence-provable main-module work this session is done + green. Remaining known issues are ALL operator/research/design-gated (cannot be closed with rock-solid proof autonomously):

- **Operator (credentials/balance):** add OPENAI_API_KEY + ANTHROPIC_API_KEY (absent); refresh GEMINI_API_KEY (live /models = "API Key not found"); recharge/replace ZHIPU_API_KEY (balance 1113). Each unblocks a verify-then-fix allowlist/E2E pass.

- **Research (needs funded keys):** ~30 other provider allowlists not audited (audit pattern proven: live /models → verify-translate + string-content shape → additive update).
- **Design:** -chunk-size CLI flag is inert (real chunking is automatic+correct via translateWithRetry/splitText; wiring the flag has ambiguous semantics OR removing it needs §11.4.122 operator confirm); markdown not a first-class CLI input (works as TXT today); reasoning-model structured-list content support in the OpenAI-compatible clients (would enable magistral/glm-5-class models).
- **Release:** NO tag yet — needs §11.4.40 full retest + operator confirm + §11.4.151 <prefix>-<version> naming.

Loop status (§11.4.94(A)/§11.4.101/§11.4.126): idle-on-blocked — every remaining item is externally/operator-gated; no item is autonomously closable with rock-solid proof. Build is the most stable this session: full sweep 54 ok/0 FAIL, websocket race-clean, format matrix complete (6 in × 5 out), deepseek current.

Session 2026-06-14o — latent brittle test-harness defects fixed (HEAD 817b9dd); full sweep GREEN 54/0

Two latent test defects (NOT caused by this session's source changes — different packages) surfaced one-per-sweep by the §11.4.118 discovery-pressure of repeated full sweeps, root-caused (§11.4.102) + fixed + proven:

- **ab7db0e** tests: DETERMINISTIC goroutine panic in tests/websocket_monitoring_test.go — concurrent map WRITE at the handleClientMessages defer delete. go test -race confirmed 22 races (suite.server.Clients + suite.testSessions + *TestSession fields accessed from the main goroutine, /status handler, and per-client goroutines without consistent locking; concurrent map read+write is a Go FATAL). Fixed: all shared-map access under suite.server.mu; thread-safe read helpers (clientCount/sessionCount/sessionSnapshot/totalSessionEvents, lock+copy, no I/O under lock) for /status + every assertion; TestMultipleClients writers joined via sync.WaitGroup (sleep = no happens-before edge → leaked writer raced another test's write to the same gorilla conn). PROVEN: -race -count=3 → 0 races; plain -count=5 → ok.
- **817b9dd** tests: DETERMINISTIC fail in cmd/translate-ssh TestSSHErrorHandling/connection_timeout — the 'connection timeout' case asserted the error contains 'connection refused', but this host's slow localhost resolver returns 'lookup localhost: i/o timeout' (§11.4.1 FAIL-bluff: failed on environment, not product defect). Broadened the connection-class assertion to accept any connection-failure indicator (refused/timeout/no route/unreachable/no such host/lookup); auth case keeps its specific check; still requires a real error. PROVEN: -count=5 → ok.

Scope note (§11.4.28): the same brittle 'connection refused' assertion class exists in the challenges/containers/helix_qa SUBMODULES (separate go.mod → NOT in the main go test ./... sweep; their own repos/sweeps — separate work, noted for follow-up).

FULL SWEEP GREEN at HEAD 817b9dd: go test ./... -p 1 = **54 ok / 0 FAIL** (qa-results/full_sweep_20260614_155722.log). build+vet exit 0. All 7 main-stream commits this session (PDF, output-format, .html, doc-sync, deepseek-v4, websocket-race, ssh-test) pushed FF to both upstreams (no force §11.4.113).

Session 2026-06-14n — provider model-allowlist §11.4.150 audit (live /models, not memory)

- **0fd1a34** deepseek: ValidModels[ProviderDeepSeek] was stale — only legacy {deepseek-chat,deepseek-coder}; deepseek.go HARD-REJECTS unlisted models, so the CURRENT flagship was rejected. Live DeepSeek /models (verified 2026-06-14) returns **deepseek-v4-flash + deepseek-v4-pro**; v4-flash proven to translate via full CLI (docs/qa/e2e_deepseek_v4_*, 136 Cyrillic chars). Added both (NOT deepseek-reasoner — not in /models, §11.4.6); kept legacy aliases (§11.4.122). RED-proven guard deepseek_v4_models_test.go. **2d98509** doc-sync: generateOutput comment now lists .html/.htm.
- **AUDIT FINDINGS (captured, for operator/future work):**
 - **Mistral:** allowlist {large,medium,small}-latest all present in live /models → FUNCTIONAL (aliases track current). NOT widened: the magistral-* reasoning family returns content as a STRUCTURED LIST (thinking blocks), which our OpenAI-compatible string-content clients CANNOT parse — adding magistral-medium would break for users. **Latent gap (operator/design):** supporting reasoning-model structured content is a non-trivial client change, not a blind allowlist add. (deepseek-v4-flash is safe because it returns content as a plain STRING with reasoning in a separate field — verified.)
 - **Gemini:** live /models re-confirms API Key not found → §11.4.6 the gemini failures are an OPERATOR CREDENTIAL issue, NOT a code bug. Operator action: refresh GEMINI_API_KEY.
 - Other ~30 providers' allowlists: not audited (no keys / would need per-provider live verification). Adding current models is safe-but-must-be-verified-per-model (string-content shape + real translation), never from memory (§11.4.99/§11.4.150).

Session 2026-06-14m — .html output added; input×output format matrix complete (HEAD fb07a59)

- **fb07a59** cli: added .html/.htm output to generateOutput (was: errored, while INPUT fully supports HTML — asymmetric gap). Minimal valid HTML5, HTML-escaped content (no markup injection). REAL E2E PROVEN (docs/qa/e2e_html_output_20260614_152301/): English PDF → real DeepSeek → out.html (valid HTML document, DOCTYPE + 2 <p>, 140 Cyrillic chars, not a zip). Guard extended (output_format_test.go): well-formed-HTML + escaping (no <script> injection) subtests.

FORMAT MATRIX NOW COMPLETE: input FB2/EPUB/TXT/HTML/DOCX/PDF (6) × output EPUB/FB2/TXT/MD/HTML (5) — all real-translation proven this session set (PDF/DOCX revived from license-gated-dead; output formats honored; HTML output added). **SESSION GRAND TOTAL: ~84 real bugs/features + 5 reconciliations.** build+vet exit 0; full go test ./... -p 1 sweep GREEN at HEAD fb07a59: **54 ok / 0 FAIL** (qa-results/full_sweep_20260614_152400.log).

Session 2026-06-14l — CLI honors -o output format + 2 §11.4.120 reconciliations (HEAD ca5608f)

- **ca5608f** cli: unified-translator ALWAYS emitted EPUB regardless of the -o extension → -o book.txt/-o book.fb2 wrote EPUB (PK-zip) bytes into a

misnamed file (silent wrong-output, §11.4). New generateOutput() dispatches on extension: .epub (default)→EPUB; .txt/.md→translated text direct; .fb2→FB2Writer; unsupported→explicit error (§11.4.6). REAL E2E PROVEN (docs/qa/e2e_output_formats_20260614_151507/): same PDF → real DeepSeek → out.txt (UTF-8 plain, not zip, 135 Cyrillic) + out.fb2 (valid FictionBook XML, 138 Cyrillic). Permanent guard cmd/unified-translator/output_format_test.go, MUTATION-PROVEN (force always-EPUB → .txt/.fb2 FAIL).

- **§11.4.120 reconciliations surfaced by the post-PDF full sweep:**

- test/unit/format_detector_test.go: PDF + DOCX moved to supportedFormats (real extractors landed); MOBI stays unsupported.
- cmd/cli TestTranslateEbookFunction/with_app_config: was a §11.4.98 BLUFF — skip-guard fired only with NO key, then forced openai + fake "config-key", dialled REAL OpenAI, 401'd, failed NoError whenever ANY other provider key was in env (DEEPSEEK/ZHIPU/GEMINI/MISTRAL present this session). Rewritten to a self-driving httptest OpenAI mock (config BaseURL→mock); asserts success + mock-hit + non-empty output. No real network, no env dependency.

SESSION GRAND TOTAL: ~83 real bugs + 5 reconciliations + §11.4.98 conversions + real E2E proofs (PDF, output-formats, 5-format matrix via DeepSeek + Mistral). All 6 input formats + 4 output formats (.epub/.fb2/.txt/.md) proven end-to-end. Post-fix full go test ./... -p 1 sweep GREEN at HEAD ca5608f: **54 ok / 0 FAIL** (was 52 ok/2 stale-gate FAIL pre-reconcile; qa-results/full_sweep_20260614_151656.log). build+vet exit 0.

Session 2026-06-14k — PDF input revived: license-gated unipdf → MIT ledongthuc/pdf (HEAD 18d6f73)

- **18d6f73** ebook: PDF INPUT was non-functional — github.com/unidoc/unipdf's ExtractText is LICENSE-GATED ("unipdf license code required" for every real PDF), so PDF input shipped dead (same §11.4 "ships but cannot be used" class DOCX was, now both fixed). Swapped to MIT github.com/ledongthuc/pdf; dropped unidoc/unipdf via go mod tidy. Wired FormatPDF into pkg/format/detector.go IsSupported + GetSupportedFormats (§11.4.108 source→artifact gate; §11.4.120 reconcile of detector_test + pdf_parser_test's unipdf-coupled error-string assertion). REAL E2E PROVEN (docs/qa/e2e_pdf_input_20260614_150421/): minimal English PDF → real DeepSeek deepseek-chat (1.9s) → valid Serbian Cyrillic EPUB ("Храбри витез жао је преко тихог сванућа у зору"), 138 Cyrillic chars, differs from source, no placeholders, no API-key leak (§11.4.10). Permanent regression guard (§11.4.135): pkg/ebook/pdf_extraction_regression_test.go + testdata/sample_text.pdf assert real extraction + reject any license-error string; mutation reverting to unipdf → FAIL.

SESSION GRAND TOTAL: ~82 real bugs + 3 reconciliations + §11.4.98 conversion + real E2E proofs (PDF + 5-format matrix via DeepSeek + Mistral). PDF + DOCX (the two dead input formats) now both work end-to-end with rock-solid real-translation evidence. ALL 6 input formats (FB2/EPUB/TXT/HTML/DOCX/PDF) proven. Post-fix targeted tests green (pkg/ebook + pkg/format); full -p 1 sweep running in background (FD ~95%, host-safe). build exit 0.

REMAINING KNOWN ISSUES (operator/research/design-gated): OpenAI/Anthropic keys absent; GEMINI_API_KEY invalid; zhipu account/model access; markdown not a first-class CLI input; other ~30 providers' model-allowlists not audited; verifier 3-tier merge precedence + run.go registry flags (design); RefreshToken no expiry re-validate (no caller); CLI always emits EPUB (no TXT writer); SQLite encryption DSN needs SQLCipher; G1/G2 carry-overs; W16 Models→submodule. NO release tag yet (needs §11.4.40 full retest + operator confirm).

Session 2026-06-14j — design/known-issue cleanup (HEAD b83038c) — 3 more real fixes + gemini/zhipu clients cleared

Subagent-driven wave (1 completed via subagent, 3 rate-limited → finished inline per §11.4.101/§11.4.147):

- **77a0c15** coordination: TranslateWithConsensus tie-break was non-deterministic (map-iteration order on vote ties) → sort candidates, strict '>', lexicographic tie-break. Mutation-proven (1000 runs identical).
- **9640a00** verification: multipass chapter loop panicked (index out of range) when the translation has FEWER chapters than the original → bound the loop by min(original,current,polished) + log/skip the tail. FACT code-analysis root cause; build+vet+tests green.
- **b83038c** ebook: FB2 writer dropped section/subsection TITLES (content was already lossless) → prepend each non-empty title as a paragraph. RED→GREEN→mutation-proven (real write→read).
- gemini/zhipu CLIENT investigation (issue #5) **RESOLVED** as **no client bug** (§11.4.6): gemini.go correctly uses <https://generativelanguage.googleapis.com/v1beta + ?key= + :generateContent>; zhipu.go correctly uses <https://open.bigmodel.cn/api/paas/v4 + /chat/completions> + Bearer. Their live failures are OPERATOR credential/account issues (invalid GEMINI_API_KEY; zhipu account lacks the model), NOT code.

SESSION GRAND TOTAL: ~81 real bugs + 3 reconciliations + §11.4.98 conversion + real E2E proofs (5-format matrix via DeepSeek + Mistral). Post-wave sweep at HEAD b83038c: 53 ok; the single pkg/distributed FAIL was ENFILE host-exhaustion ("too many open files", FD ~95%), NOT a regression — TestSSHConnection_ExecuteCommand passes deterministically in isolation (-count=3 green); nothing in pkg/distributed changed this round. build+vet exit 0.

REMAINING KNOWN ISSUES (operator/research/design-gated — see the operator-facing status report): OpenAI/Anthropic keys absent; GEMINI_API_KEY invalid; zhipu account/model access; PDF input non-functional (license-gated unipdf, same class DOCX was — needs a free PDF lib / go.mod decision); markdown not a first-class CLI input; other ~30 providers' model-allowlists not audited; verifier 3-tier merge precedence + run.go registry flags (design); RefreshToken no expiry re-validate (no caller); CLI always emits EPUB (no TXT writer); SQLite encryption DSN needs SQLCipher; G1/G2 carry-overs; W16 Models→submodule. NO release tag yet (needs §11.4.40 full retest + operator confirm).

Session 2026-06-14i — real provider×format E2E matrix found + fixed more bugs (HEAD b23bcac)

Driving REAL translations across formats/providers (rate-limited subagents → done inline) surfaced + fixed:

- **41063cb** FB2 + EPUB INPUT broken end-to-end: convertToMarkdown re-parsed already-extracted text as the original format ('failed to parse FB2: EOF' / 'not a valid zip'). Fixed → all 5 input formats translate. PROVEN: real DeepSeek E2E matrix FB2/EPUB/TXT/DOCX/HTML all → valid Serbian-Cyrillic EPUB (docs/qa/e2e_format_matrix_*).
- **9821e0f** zhipu missing from cmd/unified-translator resolveProviderAPIKey env map → ZHIPU_API_KEY never satisfied the gate. Fixed (proven: now reaches the live Zhipu API). + Multi-provider proven: Mistral E2E PASS (docs/qa/e2e_multiprovider_*).
- **b23bcac** stale provider model-allowlists (gemini/groq/zhipu accepted only DEPRECATED models, rejected current → providers unusable). Added current models; live-proven the gate blocker is gone (groq llama-3.3-70b → 429 'recognized'; gemini-1.5-flash → reaches API (auth err); zhipu glm-4-flash → reaches API (account err)). docs/qa/allowlist_fix_*.

SESSION GRAND TOTAL: ~78 real bugs + 3 reconciliations + §11.4.98 conversion + real E2E proofs (5-format matrix via DeepSeek + Mistral 2nd-provider). OPERATOR/research-gated remainders: add OPENAI/ANTHROPIC keys (absent); GEMINI_API_KEY appears invalid/expired (gemini reaches API but 'API Key not found'); zhipu key's account/endpoint rejects documented models (possible gemini/zhipu CLIENT endpoint bug like Qwen's — flagged for investigation, not guessed §11.4.6). Post-wave full sweep GREEN at HEAD b23bcac: 54 ok / 0 FAIL. Build+vet exit 0.

Session 2026-06-14h — operator-selected directions ALL executed (HEAD c6c2930); real E2E translation PROVEN

Operator selected all four next-directions; executed with conductor verification + push FF:

- **Qwen endpoint** (556002d + 2f0e359): client posted native DashScope path under a compatible-mode base AND the CODE DEFAULT base was native /api/v1 while the struct+config are compatible-mode → wrong URL + unparseable shape (Qwen was broken for default+production). Fixed: default base → /compatible-mode/v1 + path /chat/completions + choices[] (all consistent, §11.4.150 docs-verified). Live test made §11.4.98-honest (SKIP on auth/endpoint failure).
- **Storage dedup** (367adce): cache served STALE translation on tuple collision (no UNIQUE index; dedup only on id PK) → lookup_hash UNIQUE index + idempotent UPSERT + safe dedup migration, both backends. Mutation-proven.
- **DOCX** (d433210): input was NON-FUNCTIONAL (license-gated unioffice returned 'license required' for every real .docx). Rewrote as a stdlib parser (archive/zip + encoding/xml over word/document.xml + docProps/core.xml); dropped unioffice (+ transitive msoleps) via go mod tidy. Proven on a real in-test .docx (text+metadata+UTF-8). DOCX input now actually works.

- **Stress/chaos + metrics race** (bf81517): added §11.4.85 stress/chaos suites (events/coordination/distributed) which UNCOVERED a real VersionManager.metrics data race + 40-60% lost counter updates → fixed with metricsMu + snapshot-on-read; KNOWN_BUG guard flipped to active GREEN (§11.4.115).
- **CLI env-key gate** (c6c2930): unified-translator's API-key gate ignored provider * _API_KEY env vars → fixed (falls back to resolveProviderAPIKey).
- **REAL E2E TRANSLATION PROOF** (c6c2930, docs/qa/e2e_deepseek_20260614T130829/): unified-translator translated test/assets/crow_and_pitcher_en.txt EN→Serbian via REAL DeepSeek (deepseek-chat, 2.89s) into a VALID EPUB — 205 Cyrillic chars ('Гавран и крчаг...'), differs from source, no placeholders, no key leak (§11.4.83/§11.4.107/§11.4.10). Full detect→parse→translate→EPUB-write pipeline confirmed working after the session's fixes.
- **DOCX CLI wiring** (6304f60): a broader multi-format E2E proof caught that the CLI still rejected .docx with 'format docx is not yet supported' — format.Detector.IsSupported/GetSupportedFormats excluded FormatDOCX (§11.4.108 source→artifact gap: parser fixed+registered but the pipeline gate refused it). Added FormatDOCX to both; reconciled detector_test (§11.4.120). REAL E2E proven (docs/qa/e2e_docx_pipeline_20260614T131850/): real .docx → real DeepSeek → Serbian Cyrillic EPUB. DOCX input now works through the actual CLI.

SESSION GRAND TOTAL: ~75 real bugs + 3 reconciliations + §11.4.98 conversion + 2 real E2E proofs (DeepSeek on TXT + on DOCX, both → valid Serbian-Cyrillic EPUB). OPERATOR ACTIONS to widen coverage: add OPENAI_API_KEY / ANTHROPIC_API_KEY to ~/api_keys.sh (not present; ~34 other providers available). FLAGGED (not fixed, needs current-model research §11.4.150): the gemini provider's valid-model allowlist is STALE — accepts only gemini-pro/gemini-pro-vision (both deprecated by Google), rejects gemini-2.0-flash/1.5. Post-wave full sweep GREEN at HEAD 6304f60: 54 ok / 0 FAIL. Build+vet exit 0.

Session 2026-06-14g — 7th/final §11.4.118 second-pass wave (HEAD 5dfe003) — 4 more real bugs; whole-codebase second pass essentially COMPLETE

- **b1b747f** progress: formatDuration trailing space in dashboard ETA ('5 hours '); events: generateEventID non-unique within a microsecond (8172 dups/10000 in a burst → broke ID-keyed consumers). Both fixed.
- **5dfe003** cmd: preparation-translator -providers flag silently ignored (hardcoded list); cli explicit -provider silently overridden by config default (precedence inversion). Both fixed.
- **CLEAN second-pass** (honest §11.4.6, no fabricated bugs): logger, version, hardware, report, format, websocket, script.

SESSION GRAND TOTAL: 68 real bugs (5 parallel waves + 3 inline + 7 second-pass waves) + 2 reconciliations + §11.4.98 conversion. ALL reproduce-first + mutation-proven + conductor-reverified (-race/-p 1) + pushed FF (no force §11.4.113). 3 distinct security issues (CWE-204 enumeration, CWE-22 path traversal, CWE-208 timing oracle); extensive data-loss/correctness across ebook

pipeline (EPUB/FB2/HTML/markdown round-trips), providers (Anthropic/OpenRouter/Qwen), distributed, storage/cache, verification, preparation, translator-core, batch, coordination, grpc, api. The codebase has had a comprehensive 2-pass hunt; remaining items are operator/review-gated (see flags below). Post-wave full sweep GREEN at HEAD 5dfe003: 54 ok / 0 FAIL (fullsweep_20260614T120039.log). Build+vet exit 0.

OPERATOR/REVIEW-GATED carry-overs (NOT autonomously fixable — need a decision): Qwen native-vs-compatible endpoint mode; storage cache dup-tuple UNIQUE index (schema/dedup-semantics); DOCX unioffice license (go.mod dependency); detected source-lang not propagated to llm translator (cross-package wiring); FB2 writer section-title flatten; verifier 3-tier merge precedence + run.go registry flags; coordination consensus tie-break determinism; G1 verify→server-DB bridge; G2 batched ~30-provider sweep; G3 add OPENAI/ANTHROPIC keys.

Session 2026-06-14f — 6th §11.4.118 second-pass wave (HEAD 837f528) — 4 more real bugs

- **53b5056** batch: directory batch to a non-existent output dir collided ALL files onto one path (only last survived) — data loss; fix: extensionless non-existent OutputPath treated as a directory.
- **df8a8cb** coordination: TranslateWithRetry ignored ctx cancellation (full sweep+sleep on cancelled ctx); TranslateWithConsensus only scanned the first requiredAgreement instances → skipped available-and-agreeing instances behind unavailable ones. Both fixed.
- **837f528** translator: successful-but-empty translation ("", nil) silently OVERWROTE source content across metadata/chapter/section/subsection (7 fields) — 'tests pass but book destroyed' data loss; fix: keepIfTranslationEmpty preserves source when translation empty + source non-empty. Latent flag: detected source-lang not propagated to llm translator.

SESSION GRAND TOTAL: 64 real bugs (5 parallel waves + 3 inline + 6 second-pass waves) + 2 reconciliations + §11.4.98 conversion. Post-wave full sweep GREEN at HEAD 837f528: 54 ok / 0 FAIL (fullsweep_20260614T114733.log).

Session 2026-06-14e — 5th §11.4.118 second-pass wave (HEAD 367d88b) — 5 more real bugs (grpc/security/preparation)

- **45f6152** grpc: GetTranslationStatus returned codes.Unknown for unknown session (→ NotFound); cleanupOldSessions leaked session timeout context (didn't call CancelFunc → 24h timer/goroutine leak).
- **5871848** security: CWE-208 username-enumeration TIMING oracle — no bcrypt compare on user-not-found path → ~1.7M× faster than wrong-password → reveals valid usernames. Fix: dummy-bcrypt compare at init (OWASP constant-time).
- **367d88b** preparation: failed-chapter analysis MISATTRIBUTION (positional index after results compaction → chapter got wrong chapter's analysis; fix: lookup by ChapterNum); nested-subsection content DROPPED from analysis input (extractors ignored Section.Subsections; FB2 populates them; fix: recursive writeSectionContent).

SESSION GRAND TOTAL: 60 real bugs (5 parallel waves + 3 inline + 5 second-pass waves) + 2 reconciliations + §11.4.98 conversion. Post-wave full sweep GREEN at HEAD 367d88b: 54 ok / 0 FAIL (fullsweep_20260614T113420.log).

Session 2026-06-14c/d — two more §11.4.118 second-pass waves (HEAD 4a95b06) — 11 more real bugs; 3-stream low-FD method proven repeatable at ~95% FD, NO ENFILE

Wave c (verification/storage/markdown) = 7 bugs; wave d (api/verifier/script) = 4 bugs + script came back genuinely CLEAN (honest §11.4.6, no fabricated bug). All conductor-reverified (build+vet+ -p 1 per pkg + full sweep -p 1) and pushed FF.

- **34e09f8** verification: parseNote dropped a note ending in EXAMPLES (no IMPLICATIONS); NaN Confidence (count==0 divide); multipass section under-count (pointer aliasing).
- **9946d85** storage: Redis makeCacheKey metadata delimiter-injection collision (Ollama 'llama3:8b' has ':') → wrong translation; fixed NUL-joined hash.
- **d055b8c** markdown: EPUB[?]MD code blocks destroyed, GFM tables lost both ways, backslash-escapes corrupted — all fixed bidirectional.
- **b8bc3ec** api: translateText silently dropped script=cyrillic + accepted invalid script (→ validate+switch); verifier handlers mapped all GetModel errors to wrong status (→ errors.As 404-vs-503).
- **4a95b06** verifier: buildFallbackChain rand.Shuffle destroyed score order + nondeterministic (→ keep FilterVerified order); OpenRouter pricing typed float64 but live API emits quoted strings → whole Tier-2 decode failed → 0 OpenRouter models (→ json.Number).

Honest flags (NOT fixed — ambiguous/spec-gated): verification polishWithNotes equal-chapter-count assumption; verifier run.go registry CanSeeCode/AffirmativeResponse + 3-tier merge precedence; api getStats nil-cache (not prod-reachable) + distributedManager-gated SSRF/null-JSON (integration-gated); storage dup-tuple unique-index (review-gated); script Convert short-circuit leaves stray off-script chars (deliberate pinned optimization). pkg/script second-pass = CLEAN (mature, all 30 letters map, digraph casing correct).

SESSION GRAND TOTAL: 55 real bugs (5 parallel waves + 3 inline + 4 second-pass waves) + 2 reconciliations + §11.4.98 conversion. Post-wave full sweep GREEN at HEAD 4a95b06: go test ./... -p 1 = 54 ok / 0 FAIL (fullsweep_20260614T112302.log). build+vet exit 0.

Session 2026-06-14b — parallel §11.4.118 second-pass discovery wave (HEAD 47ea69d) — 5 more real bugs; ran 3 subagents at ~95% FD with low-FD discipline, NO ENFILE

Operator repeatedly requested parallel subagents; FDs never freed (~95%, 15 claude procs). Honored the informed repeated instruction with mitigations: 3 streams (not 4), investigation-first + targeted -p 1 tests only, no ./... , no -race during hunt. Result: NO ENFILE, 5 real bugs, conductor-reverified (build+vet+ -p 1 each pkg green), 3 commits pushed FF.

- **87a8048** — Anthropic client returned only response.Content[0].Text but the Messages API returns content as an ARRAY: long output split across text blocks was TRUNCATED, and a leading thinking/redacted_thinking block made

Content[0].Text=="" → EMPTY translation. Fix: concatenate all text blocks (skip thinking/tool). httptest RED→GREEN→mutation.

- **edadd73** — distributed performance.go ResultCache.Set evicted on every update-while-full (didn't check key-exists) → shrank cache below maxSize, dropped valid entry. + coordinator.go round-robin currentIndex never reset on DiscoverRemoteInstances rebuild → index-out-of-range PANIC after worker pool shrinks (translation hot path). Both RED→GREEN→mutation (-p 1).
- **47ea69d** — ebook epub_parser.go built opfDir+href verbatim; EPUB OCF hrefs are percent-encoded/relative (RFC 3986) but zip entries are literal → 'chapter%20one.xhtml' never matched 'OEBPS/chapter one.xhtml' → chapter+cover SILENTLY DROPPED (data loss). Fix: resolveEPUBHref (PathUnescape + ./ ./ normalize). + html_parser.go removed a hardcoded 'Nestedtexthere'→'Nested text here' test-hack that corrupted any real doc containing that substring. RED→GREEN→mutation.

Honest §11.4.6 flags (NOT auto-fixed — ambiguous intent/operator-gated): Qwen endpoint² response-shape mismatch (native DashScope path vs OpenAI-compatible config base-URL — needs operator decision on intended Qwen mode); coordinator reduced_quality fallback lowercases passthrough text; FB2 writer section-title flatten. Markdown→EPUB path clean-verified (lists/blockquotes/fences/headers all handled). EPUB writer clean-verified.

SESSION GRAND TOTAL: 44 real bugs (5 parallel waves + 3 inline + this 5-bug second-pass wave) + 2 reconciliations + §11.4.98 conversion. Post-wave full sweep GREEN at HEAD 47ea69d: `go test ./... -p 1 = 54 ok / 0 FAIL` (host-safe -p 1 avoided ENFILE at ~95% FD); evidence qa-results/full_sweep/fullsweep_20260614T104630.log. Build+vet exit 0.

Purpose: Single canonical out-of-the-box entry point for any fresh session (§11.4.131 / §12.10 / §11.4.127). To resume: point a new session at THIS file, run `git fetch --all`, and say **continue**.

Session 2026-06-14a — host-safe single-stream inline (HEAD bd8a1ef) — 2 more real data/correctness bugs in pkg/translator/llm (the chunking path subagents skipped)

Context: operator returned, asked for 3-4 parallel subagents; chose "free FDs first" but host kern.num_files stayed ~94-95% of maxfiles (13+ concurrent claude processes, NOT freed). Per §12 host-safety I did NOT launch parallel subagents (ENFILE risk); instead ran host-safe single-stream INLINE work (targeted -p 1 tests never hit ENFILE all night) and armed a background watcher (b3qgjr30s) to auto-launch the parallel wave the instant FD<80%. 2 commits, pushed FF.

- **24b0fd0** — splitText DROPPED PARAGRAPH SEPARATORS at chunk boundaries (the >20KB size-error retry path subagents had skipped as out-of-scope). It stripped "\n\n" via Split + re-added inconsistently (never around an oversized para, never across a chunk seam); reassembly strings.Join(chunks,"") then glued the last paragraph of one chunk to the first of the next; "\n\n\n\n" collapsed to "\n\n". Structural data loss in translated large chapters. Fix: splitText now tiles losslessly — re-attach each para's "\n\n" so Join(splitText(text),"")==text (makes reassembly correct by construction). RED (5 cases dropped \n\n) -> GREEN -> mutation polarity proven.

- **bd8a1ef** — in-memory cache-key COLLISION served wrong translation: key was `fmt.Sprintf("%s:%s",text,contextStr); ":"` inside text means ("a:b","c") and ("a","b:c") both -> "aBc" -> 2nd request served 1st's translation (same class as the storage Redis fix). Fix: makeLLMCacheKey injective via length prefix "%d:%s:%s". RED (Translate served `xlate(a:b)` for input "a") -> GREEN -> mutation-proven (helper body).
- **71babe1** — FB2 writer DROPPED deeply-nested section content (>2 levels): chapter loop only descended `sec.Content` + direct `sub.Content`, never `sub.Subsections` (depth 3+) -> deep translated text vanished from the .fb2 (EPUB writer's recursive `formatSection` preserves it). Fix: `collectSectionParagraphs` recurses the full `Subsections` tree. RED (real write->read: LEVEL_3/_4 missing) -> GREEN -> mutation-proven. Honest note (§11.4.6, NOT fixed — ambiguous intent): FB2 writer still flattens section/subsection TITLES to chapter level (`sec.Title` not written); content now lossless.
- EPUB writer (`pkg/ebook/epub_writer.go`) §11.4.6 clean-verified: `escapeXML` correct+complete, OPF/NCX/chapter filenames consistent, shared identifier, Chapter has no orphan `Content` field — no data-loss; no bug invented.

SESSION GRAND TOTAL now: 39 real bugs (waves 1-5 = 32 + wave-5 4... see below) + 2 reconciliations + §11.4.98 conversion. Build STABLE GREEN, go build ./... exit 0; pkg/translator/llm passes offline. Watcher b3qgr30s armed (40min cap) → auto-launch 3-4 parallel subagents for a §11.4.118 second-pass discovery wave the moment FD<80%. While FD-pressured, conductor continues host-safe single-stream inline bug-hunting.

Session 2026-06-13g — bug-hunt wave 5 (HEAD ceab1db) — 4 real bugs incl. CWE-22 path traversal + PairingManager race; build STABLE GREEN; paced for §12 host-safety

Mode: §11.4.126 autonomous loop + §11.4.103 3 parallel subagents (PairingManager mutex · pkg/language · pkg/api) + conductor verify-then-commit. 3 commits (38300e4..ceab1db) pushed FF.

- **38300e4** — pkg/distributed PairingManager.services map DATA RACE (no manager mutex; map + *RemoteService.Status/LastSeen/PairedAt written by health goroutines, read by GetPairedServices) → one RWMutex, deadlock-averse (unlocked-internal `addService` helper, no re-entrant lock, lock never held across HTTP probe / event Publish; performHealthChecks snapshots under RLock then releases). RED(-race)→GREEN(-race -count=5 ~4s no hang)→mutation; full pkg -race ok ~52s. Resolves the wave-4 flagged latent.
- **5c1a2ae** — pkg/language 2 UTF-8 bugs: `detectHeuristic` sampled text[:1000] by BYTE → under-counts trailing multi-byte script → majority-Cyrillic book mis-detected as English (→ 1000 RUNES); `FormatLanguageCode` `code[:2]` by BYTE → invalid UTF-8 on multi-byte char (→ 2 RUNES); + `DetectLanguage` prompt sample rune-fix (source-only, network-gated). Mutation-proven, -race ×3.
- **ceab1db** — pkg/api PATH TRAVERSAL (CWE-22): `uploadUpdate/applyUpdate` interpolated the X-Update-Version header into a `filepath.Join, '..'` escaped the update dir → arbitrary file write. → `validateUpdateVersion` allowlist (reject `"'"/.*/separators/NUL/>128, charset [A-Za-z0-9.+_-]`). RED(200+escaped path)→GREEN(400)→mutation; existing 11 subtests pass, -race ×3.

BUILD STATE — STABLE GREEN. Every package passes: the post-wave-5 full sweep showed 52 ok + 2 transient FAILs (pkg/translator TestMemoryUsage + test/integration build) that were proven ENVIRONMENTAL (§11.4.7), NOT code: error was literally "too many open files in system"; `go test -c ./test/integration` compiles exit 0; both re-ran GREEN at `-p 1`. Root cause: host `kern.num_files` at ~95.8% of `kern.maxfiles` (122880) from 5 waves of parallel subagents + repeated full `go test ./...` + a concurrent claude security-review process. Last fully-clean authoritative sweep: HEAD de25a98 = 54 ok / 0 FAIL (qa-results/full_sweep/fullsweep_20260613T230845.log). `go build ./...` exit 0 throughout.

§12 HOST-SAFETY DECISION (§11.4.101 safest/most-stable): host file table near saturation → STOP spawning more parallel subagents / full-repo builds until it recovers (adding load risks tipping past maxfiles = other workloads fail to open files = the destabilization §12 forbids). The autonomous-safe disjoint package queue is also essentially EXHAUSTED — the whole codebase surface has been hunted across 5 waves (llm, websocket, coordination, fb2, ebook[epub/html/docx], batch, cmd, verifier, preparation, storage, report, markdown, format, grpc, security, hardware, version, progress, services, config, cache, translator-core, distributed, language, api). Per §11.4.94(A) the remaining items are genuinely blocked → paced re-verification, not idle-by-laziness.

SESSION GRAND TOTAL (2026-06-13 → 06-14): 36 real bugs fixed across 5 waves (7+10+7+8+4) + 2 reconciliations + §11.4.98 llm offline conversion

All reproduce-first + mutation-proven + conductor-independently-reverified (-race) + 24 descriptive commits pushed FF to milos85vasic/Translator + HelixDevelopment/HelixTranslate (no force §11.4.113). Notable: CWE-204 auth enumeration oracle, CWE-22 path traversal, gRPC remote-DoS nil panic, websocket cross-session event leak, FB2/markdown/ebook round-trip data loss, cache+pairing+versionCache+ssh data races, UTF-8 mid-rune corruption (translator+language).

Flagged latent (review/operator-gated, NOT autonomously safe overnight): storage cache dup-tuple unique-index (schema/dedup-semantics change §11.4.124); DOCX unioffice license-gated (go.mod dependency decision); security RefreshToken re-validate (no caller, defense-in-depth); G1 verify→server-DB bridge; G2 batched ~30-provider sweep; G3 add OPENAI/ANTHROPIC keys to `api_keys.sh`. These need operator decisions (user away) — correctly deferred per §11.4.101/§11.4.122.

Session 2026-06-13f — bug-hunt wave 4 + full-repo sweep GREEN (HEAD de25a98) — 8 real bugs; `go test ./...` = 54 ok / 0 FAIL offline

Mode: §11.4.126 autonomous loop + §11.4.103 4 parallel subagents (internal/services+config · internal/cache · pkg/translator core · pkg/distributed deeper) + conductor verify-then-commit + post-wave full sweep. 4 commits (eeda1ee..de25a98) pushed FF.

- **eeda1ee** — services/config 3 bugs: GetPreferences claimed sorted-by-score but did NO sort (passed only on pre-sorted fixture) → sort desc; FallbackOrder used unfiltered index → first accepted pref got 2 not 1 → contiguous post-filter rank;

config all-zero scoring weights fail-open ('total>0 &&' skipped the sum check) → reject. Mutation-proven.

- **9d4f4a4** — internal/cache cleanup goroutine LEAK: NewCache(,true) launched 'for range ticker.C' with no Close/stop → one goroutine + pinned Cache leaked per instance forever (unbounded on a long-running server). Added stop chan + Once-guarded Close() + cleanupWG; backward-compatible. Mutation-proven (200 caches→200 leaked).
- **c1bd61f** — pkg/translator universal.go language-detection sample sliced sample[:2000] by BYTE → mid-rune split → invalid UTF-8 to Detect (common case for Cyrillic/CJK ebooks) → truncateOnRuneBoundary. Mutation-proven, drives real detector.
- **de25a98** — pkg/distributed 3 bugs: checkServiceHealth set Status='online' unconditionally → every paired worker DEMOTED+dropped after first 30s tick → all work distribution lost (HIGH); versionCache map had NO mutex (existing one is AlertManager's) → BatchUpdateWorkers data race → cacheMu; ssh LastUsed two-locks-one-field race → conn.mu touch()/idleSince(). Mutation-proven, full pkg -race ok.

STABILITY MILESTONE (re-confirmed): go test ./... -count=1 all provider env unset = **54 ok / 0 FAIL**, go build ./... exit 0. Evidence: qa-results/full_sweep/fullsweep_20260613T230845.log. SESSION GRAND TOTAL: 32 real bugs (waves 1-4: 7+10+7+8) + 2 reconciliations + §11.4.98 llm offline conversion, all reproduce-first + mutation-proven + conductor-reverified (-race) + pushed FF (no force §11.4.113).

Flagged latent (honest §11.4.6, NOT autonomously fixed — review/larger-blast-radius): PairingManager.services map needs a manager-level mutex (~10 methods); storage cache dup-tuple unique-index (schema/dedup-semantics change); DOCX unioffice license-gated (go.mod); security RefreshToken re-validate (no caller). Operator-gated: G1/G2/G3.

Next-wave queue (disjoint, autonomous-safe, thinning): pkg/language, pkg/api (deeper non-integration logic), the PairingManager mutex (careful single-mutex + -race). After that the autonomous-safe disjoint queue is largely exhausted → legitimate idle per §11.4.94.

Session 2026-06-13e — bug-hunt wave 3 + full-repo sweep GREEN (HEAD 8d0b1b6) — 7 real bugs + 2 interaction-defect reconciliations; go test ./... = 54 ok / 0 FAIL offline

Mode: §11.4.126 autonomous loop + §11.4.103 4 parallel subagents (grpc · security · ebook html/docx · hardware/version/progress) + conductor verify-then-commit. Then a full-repo go test ./... sweep (§11.4.40-style interaction check) caught 2 latent defects, both reconciled. 6 commits (6e1b1f2..8d0b1b6) pushed FF.

- **6e1b1f2** — grpc 2 bugs: StartTranslation deref'd req.ProviderConfig.Type unconditionally (proto3 optional → nil panic, no grpc recovery → remote-DoS CRASH the serving goroutine) → validate up front, InvalidArgument; session-limit TOCTOU (RLock-check then Lock-insert) + duplicate SessionId silently overwrote prior session WITHOUT CancelFunc (timeout-context leak + orphaned translation) → atomic gate+reserve, AlreadyExists. Mutation-proven over bufconn, -race ×2.
- **96999b3** — security: AuthenticateUser checked IsActive BEFORE password → username/account-status enumeration oracle (CWE-204; API maps to 403 vs

401) on the live login path → validate password FIRST, disclose inactive only with correct password. Also fixed a §11.4.1 JWT-tamper test FAIL-bluff (mutated last sig char, ~1/64 flaky → first char, 200/200).

RED(exploit)→GREEN→mutation, -race ×3, downstream pkg/api green.

- **95323d8** — ebook HTML: td/th/dt/dd cells concatenated with no separator ('cell Acell B') + dropped ('line oneline two') → isCellElement newline + newline. Mutation-proven. HONEST PENDING FORENSICS: DOCX parser depends on license-gated unioffice → non-functional for real DOCX (go.mod-scope, not fixed).
- **044bacb** — hardware: vm_stat malformed page-size discarded parse error → 0 available RAM → guard (err==nil && >0); meminfo kB*1024 uint64 overflow wrap → maxKB guard. progress: GetProgress clobbered Complete()'s 'Completed' ETA → gate override on <100% && status!='completed'. Mutation-proven.
- **8d0b1b6** — full-sweep interaction reconciliations (§11.4.120/§11.4.98): test/unit scoring expectations updated to the FIXED weight formula (0.78→0.675, 0.625→0.5, math-verified, not a revert); cmd/cli TestTranslateEbookFunction/with_app_config dialed real Qwen (env-dependent) → honest §11.4.3 SKIP when no provider key.

STABILITY MILESTONE: go test ./... -count=1 with ALL provider env unset = **54 ok / 0 FAIL / 24 no-test-files**, deterministic offline. go build ./... exit 0. Evidence: qa-results/full_sweep/fullsweep_20260613T225207.log. Session total this run: 26 real bugs (waves 1+2+3) + 2 reconciliations, all reproduce-first + mutation-proven + conductor-reverified + pushed FF (no force §11.4.113).

Next-wave queue (disjoint, autonomous-safe): internal/services, internal/config, internal/cache, pkg/translator (deeper, non-llm), pkg/distributed (deeper pure-logic), cmd more. Carry-over latent (review-gated, NOT autonomously fixed): storage duplicate unique-index; DOCX unioffice license (go.mod); security RefreshToken re-validate. Operator-gated: G1/G2/G3.

Session 2026-06-13d — parallel subagent bug-hunt wave 2 (HEAD 39ee07a) — 10 real bugs, all reproduce-first + mutation-proven + conductor-reverified (-race) + pushed FF (no force §11.4.113)

Mode: §11.4.126 autonomous loop + §11.4.103 4 parallel subagents (disjoint scopes: internal/verifier · pkg/preparation · pkg/storage+report · pkg/markdown+format) + conductor verify-then-commit. 4 descriptive commits (539659b..39ee07a) pushed FF to milos85vasic/Translator + HelixDevelopment/HelixTranslate.

- **539659b** — internal/verifier 3 bugs: scoring/engine.go applied weights.Capability TWICE + DROPPED Recency (sum 1.15, broke 0..1 contract); discovery HF models keyed on legacy modelId only → valid id-only payload collided every model on ID="" → ZERO usable models; registry FilterVerified/ListModels map-iteration nondeterministic order (latent flake). All mutation-proven, -race.
- **1bee190** — pkg/preparation 3 bugs: analyzeChapters appended goroutine results in COMPLETION order → GetTranslationContext[n-1] mis-attributed each chapter's analysis to the wrong chapter (→ slot-indexed, no mutex); multi-pass consolidation set FinalAnalysis from fresh parse with no chapter_analyses → all per-chapter analysis silently lost (→ re-attach); isUntranslatable

strings.Contains(text,"")==true → blank LLM term marked EVERY string untranslatable incl. title (→ skip empty). Mutation-proven, -race ×3.

- **0a415b1** — pkg/storage negative CacheHitRate in GetStatistics (all 3 backends sqlite/postgres/redis): (totalAccess-totalTranslations)/totalAccess went negative when entries inserted-but-rarely-reread (3 entries, 1 hit=-200%) → hits/(hits+misses), always [0,100). Mutation-proven via real SQLite round-trip.
- **39ee07a** — pkg/markdown 3 reader-visible EPUB-output bugs in the PRODUCTION chapter path: h4-h6 headers shipped as literal <p>#### x</p> (only 1-3 handled; §11.4.108 source→artifact — h4-6 lived only in a dead sibling fn); fenced code blocks flattened to one <p> with literal fences+collapsed newlines (→ <pre><code>); SimpleWorkflow wrote body/header text RAW → malformed XHTML on &/</> (→ escapeXHTMLText). Mutation-proven.

Honest clean-verifications (§11.4.6): coordination/scoring/selection concurrency sound; openai/qwen/hipu/llamacpp clients parse correctly; format detector edge cases correct; Redis cache-key+avg-duration sound (prior fixes intact).

FLAGGED latent (NOT fixed — needs conductor/operator review per §11.4.124, schema change): pkg/storage SQLite/Postgres cache has NO UNIQUE index on the lookup tuple (source_text,source_language,target_language,provider,model); storing the same tuple under 2 different ids leaves 2 rows and GetCachedTranslation can return the older STALE row. No production caller constructs cache IDs today (all in tests). Correct fix = UNIQUE index migration changing dedup semantics → review-gated.

Next-wave queue (disjoint, autonomous-safe): pkg/distributed deeper (conductor-owned §11.4.119), pkg/grpc, pkg/security deeper, pkg/ebook html/docx parsers, pkg/hardware, pkg/version/progress. Carry-over: storage dup-tuple unique-index (review-gated); G1/G2/G3 operator-gated.

Session 2026-06-13c — parallel subagent bug-hunt wave (HEAD f451468) — 7 real bugs, all reproduce-first + mutation-proven + conductor-reverified (-race) + pushed FF (no force §11.4.113)

Mode: §11.4.126 autonomous loop + §11.4.103 4 parallel bug-hunt subagents (disjoint package scopes) + conductor verify-then-commit (§11.4.147/§11.4.142). Conductor independently re-ran build+vet+-race on every affected package and reviewed every production diff BEFORE committing. 6 separate descriptive commits (caa39c1..f451468), pushed FF to milos85vasic/Translator + HelixDevelopment/HelixTranslate.

- **caa39c1** — websocket Hub.StartServer SESSION FAN-OUT HOLE: /ws handler left Client.SessionID empty → hub's per-session filter skipped → every dashboard got EVERY session's events (real caller cmd/ssh-translation). Fixed (read session_id from query) + moved off global DefaultServeMux→private mux (dup-/ws panic). Mutation-proven (alpha client saw beta's BETA-LEAK frame), -race ×3.
- **91e71c4** — fb2 Paragraph had UnmarshalXML but NO MarshalXML (Text/FullText are xml:"-") → parse→write round-trip emitted empty <p></p> = TOTAL paragraph data loss. Added MarshalXML (FullParagraphText, id/style attrs, escaped once). Mutation-proven.
- **ef8b3c7** — ebook EPUB fallback cleaner 2 bugs: CleanXMLData blind 2-char prefix rewrite corrupted valid entities (&→∓) + text (Q&A);

removeHTMLTags regex EXCLUDED self-closing/void tags (

-) → leaked literal markup into chapter text. Fixed (escapeBareAmpersands + unified htmlTagRe, RE2-safe). Mutation-proven; stale test reconciled §11.4.120.
- **7e05c5e** — batch Process(ctx) NEVER honored cancellation (no ctx.Err/Done in either loop) → ran every remaining file, error stayed nil. Added seq guard + parallel in-worker short-circuit + post-Wait surfacing. Mutation-proven ×2, -race ×3.
- **3225c8d** — cmd UPPERCASE/mixed-case ext not stripped: TrimSuffix(base, ToLower(ext)) left "Story.EPUB" → "Story.EPUB_sr.epub" (case-insensitive FS). Fixed in unified-translator (3 helpers) + translator. Mutation-proven; stale test reconciled §11.4.120; cmd/translator gained first tests.
- **f451468** — §11.4.98 offline-ize: openai/qwen/zipu error-paths → httpstest; llamacpp auto-download → temp-HOME stub .gguf (no HF dial); HF_TOKEN skip-guards; TestValidate AvailableRAM pinned. Whole pkg/translator/llm now passes OFFLINE ~8s, all provider env unset, mutation-checked each converted test still has teeth. No prod client bug (honest §11.4.6).

Honest clean-verifications (§11.4.6 — investigated, NO bug): pkg/coordination (lock ordering leaf-only, consensus chan cap ≥ goroutines, immutable instances post-init); openai/qwen/zipu/llamacpp clients parse correctly (defect was purely tests dialing real APIs).

Next-session queue (disjoint, autonomous-safe): bug-hunt internal/verifier (scoring/discovery/selection/client), pkg/preparation, pkg/storage (deeper), pkg/distributed (deeper pure-logic), pkg/markdown/format/html/docx parsers deeper. Carry-over G1 (verify → server-DB bridge), G2 (batched ~30-provider sweep), G3 (operator: add OPENAI/ANTHROPIC keys to api_keys.sh).

Session 2026-06-13b — parallel subagent bug-hunt wave (HEAD 86175d0) — ~15 real bugs, all reproduce-first + mutation-proven + -race + pushed FF (no force §11.4.113)

Mode: §11.4.126 autonomous loop + §11.4.103 ≥3 parallel streams (conductor + 4–5 background bug-hunt subagents/wave). §11.4.147 exercised heavily — multiple subagents were rate-limit-crashed ("Server is temporarily limiting requests") and respawned until complete; conductor independently verified every subagent claim (-race + scope census + §11.4.142 diff review) BEFORE committing. Commits pushed to milos85vasic/Translator + HelixDevelopment/HelixTranslate, fast-forward, no force.

- **a40a9b9** — logger shouldLog fail-open (typo level "warning" → logged everything incl DEBUG in prod) + fail-drop (unknown msg level dropped) → levelRank fallback (cov 84.3 → 91.1%); progress tracker 5 bugs (GetProgress data race, Complete() 0% clobber, negative %, ETA float truncation, divide-by-zero PANIC); ollama_test 4 subtests dialed real httpbin.org → httpstest (6.6s → 0.3s offline) §11.4.98.
- **3a7ae78** — gRPC EventBus handler LEAK (SubscribeEvents never unsubscribed → unbounded allEvents growth) → added SubscriptionID+Unsubscribe+HandlerCount, defer Unsubscribe, drop close; + pre-existing session-field DATA RACE (sessionsMutex guarded only the map) → locked Status writes, snapshot-under-RLock GetTranslationStatus,

fixed latent recursive-RLock deadlock in ListTranslations. Mutation-proven (events + real bufconn leak test), -race ×3.

- **a2aa3da** — report ReportGenerator data race (unsynced appends) → sync.Mutex + snapshot-under-lock; deepseek/anthropic/gemini tests dialed real api.* → httptest (package 30.6s→19.2s offline) §11.4.98.
- **ef0a60f** — verification multipass $O(n^2)$ DUPLICATE DB WRITES (polishWithNotes re-saved the whole accumulated report.SectionResults each chapter → polishing_changes got 1+2+...+N dup rows; section_results PK-conflict inserts failed silently) → saveNewResults cursor. Mutation: 15 change rows for 5 chapters vs 5.
- **916dddd** — storage Redis cache-key HASH COLLISION (32-bit poly-31 → distinct texts shared a key → WRONG cached translation served) → sha256; + Redis avg-duration dilution → accumulateAvgDuration. Mutation-proven.
- **86175d0** — markdown→EPUB ROUND-TRIP DATA LOSS: links/images/blockquotes left literal in HTML AND the production convertMarkdownToXHTML chapter path (readers got [text](url) / >) → blockquote accumulation + image/link inline conv (image-before-link). start renumber pinned KNOWN-LOSSY. 4 bugs mutation-proven.

Honest clean-verifications (§11.4.6 — investigated, NO bug): gRPC StreamTranslationProgress (streams-map delete-under-lock + non-blocking send ⇒ no send-on-closed); preparation extractJSON (string-aware matchBalanced + json.Valid guard + fenced handling robust; crashed preparation subagent left only scratch, removed §11.4.84).

REMAINING NET §11.4.98 offenders (still dial real api.* — confirmed via keys-stripped per-test timing): **openai** (TestOpenAITranslateErrorPaths), **qwen**, **llamacpp** (HF download paths), **zhipu** invalid_model. Same httptest pattern → strong next-wave item.

Next-session queue (disjoint, autonomous-safe): finish NET httptest-ization (openai/qwen/zhipu/llamacpp); bug-hunt cmd/* entypoints, pkg/websocket, pkg/coordination, pkg/batch, pkg/fb2, pkg/ebook parsers/writers, pkg/distributed deeper, internal/verifier. Carry-over G1 (verify→server-DB bridge), G2 (batched ~30-provider sweep), G3 (operator: add OPENAI/ANTHROPIC keys to api_keys.sh).

Session wrap-up — W13, W15 (api/grpc), W16 all resolved (HEAD 1cc66ac)

- **W15 api/grpc DONE (real, evidence-backed)** — subagent-driven. Built build-tagged integration suites: pkg/api/server_realhttp_integration_test.go (real HTTP via httptest + real JWT + real Postgres via brokertest: health, JWT 401/200, login→token round-trip, /api/v1/verified-models, session persist round-trip) and pkg/grpc/server_storage_integration_test.go (real gRPC over bufconn + real Postgres: submit→persist, status, cancel, list-count, 8× concurrent, error-code via status.FromError; §1.1 mutation-proven). LLM-translate leg = honest §11.4.3 SKIP (no provider key). Normal go test boots NOTHING; SKIP-clean without podman.
- **REAL AUTH BUG fixed (W15 api leg, commit 0e2a999)** — InMemoryUserRepository.Create stored the caller's pointer; CreateUser clears Password="" post-Create → blanked the stored bcrypt hash → EVERY user registered via CreateUser could never log in. Fix: repo stores a COPY. Conductor-reproduced RED(401)→GREEN(200) on

TestAPIRealHTTP_LoginTokenRoundTrip (§11.4.115). Remaining W15: only the pure LLM-translation E2E (operator-gated on a real provider key).

- **W13 DONE (commit 1cc66ac)** — 8 PascalCase .gitmodules section labels → lowercase (paths were already compliant). Done §9.2 backup-first; all 11 submodules verified resolving to identical baseline SHAs after git submodule sync, dirty challenges/helix_qa preserved, build exit 0, backup removed post-gate.
- **W16 RESOLVED = KEEP Models/ as-is** — the pkg/modelsbridge bridge already consumes digital.vasic.models (functional need met). Converting Models/ → a git submodule requires creating a NEW upstream repo under an owned org — an operator-gated repo-creation action (§11.4.101 block-only-when-irreversible+undeterminable; §11.4.122 keeping is the safe no-change path). Convert path remains available on operator request.

Autonomous-safe queue now EMPTY. Genuinely-remaining = operator-gated only: W15 LLM-translation E2E (real provider key) + W16 convert (new upstream repo decision). Per §11.4.94 this is legitimate idle.

LLMsVerifier real-key integration — DONE (HEAD e7d6d67; submodule 95fb1cce)

Operator directive: use ~/api_keys.sh so the LLMsVerifier submodule performs its mandatory steps and provides valid providers + models to the System. **Done, evidence-backed (§11.4.83 docs/qa/llmsverifier_20260613_182600/, key-free):**

- Ran the submodule's REAL discovery/verification with real keys → **89 verified models** (DeepSeek 2, Cerebras 2, Groq 16, Mistral 69; OpenAI/Anthropic honestly fail — no key in api_keys.sh).
- **Two real bugs found + fixed, both §11.4.135 mutation-proven guards (exact production errors):**
 - **Bug B (main, internal/verifier/client.go) load-bearing:** the System's SSOT client couldn't decode the server's {"count":N,"models":[...]} envelope → consumed ZERO models while server healthy. Fixed (envelope-aware decode). Guard: client_envelope_test.go. System now fetches all 89 e2e.
 - **Bug A (submodule llmverifier/models.go, 95fb1cce, pushed GitHub+GitLab):** numeric context_window (Groq) crashed the whole provider decode → 0 models. Fixed (tolerant UnmarshalJSON). Guard: context_window_unmarshal_test.go.
- Secrets: keys sourced inline only, never printed/logged/committed (§11.4.10); evidence dir + edits secret-scanned clean; throwaway run artifacts removed (§11.4.14).

Honest follow-up gaps (tracked, not bluffed — §11.4.6):

- (G1) Submodule verify (report) pipeline does NOT persist into the server SQLite DB — the served DB was seeded via live discovery (verifier's own client+database pkgs). Architectural bridge worth a workable item.
- (G2) Full ~30-provider exhaustive per-model coding-capability scoring timed out at 600s in one pass; reliability-scored the seeded set. A bounded/batched sweep is the follow-up.
- (G3) Providers without keys (OpenAI/Anthropic/Perplexity/Together) fail auth honestly; add their keys to api_keys.sh to include them.

Real-bug-hunt waves (HEAD eb7576e) — 17 real bugs fixed this session, all reproduce-first + mutation-proven + pushed

LLMsVerifier real-key integration: 89 verified models flow to the System end-to-end; System's own in-process verifier env-bridge (31 models, cmd/verify-models); real DeepSeek translation proof ("The sky is blue today."->"El cielo esta azul hoy.").

Bugs fixed (each RED->GREEN, §11.4.135 guard, §1.1 mutation-proven, real evidence; commits e7d6d67/dfb250c/f80153b/eb7576e + submodule 95fb1cce):

- Auth lockout (CreateUser blanked stored bcrypt hash); Verifier consume (client.go envelope); Verifier serve (listVerifiedModels dropped all); Groq context_window (submodule); Hardcoded RU->SR prompt (RU->SR Ekavica preserved); Uppercase Cyrillic digraph (LJUBAV); Ordered-list numbering lost; UTF-8 mid-rune truncation (HIGH, Cyrillic); Import data race; FB2 inline+tail text dropped (CRITICAL); FB2 writer escaped markup; FB2 format misdetection; EPUB OPF/NCX id mismatch; Markdown inline whitespace collapse; silent chapter drop; list reverse round-trip; extractJSON wrong span.

Remaining tracked follow-ups (honest, §11.4.6/§11.4.118):

- G1: submodule verify-pipeline->server-DB bridge (serial submodule wave, §11.4.119 contends with main builds; conductor-owned).
- G2: batched full ~30-provider verification sweep (one-pass timed out 600s).
- G3: add OPENAI/ANTHROPIC/etc keys to api_keys.sh (operator) to include those providers.
- Deferred-broader: markdownToHTML round-trip edges; preparation extractJSON LLM-output spec; verification multipass duplicate DB writes; ambiguous nj/lj/dz Latin->Cyrillic (structural §11.4.112 won't-fix).

Bug-hunt wave 3 (HEAD 6c76c3f) — 6 more real bugs (session total ~23), all mutation-proven + pushed

- CRITICAL OpenAIClient panic on non-float64 temperature (blast radius = ALL OpenAI-compatible providers) — toFloat64/toInt coercion.
- CLI -temperature/-max-tokens ignored by openai/anthropic/qwen/zhipeu — Options>typed-field>default precedence.
- Gemini discarded valid MAX_TOKENS partial text (data-loss) — accept STOP/MAX_TOKENS/"" (gemini_test.go reconciled §11.4.120).
- EventBus.Publish unbounded goroutine-per-handler + RLock-held-across-spawn → explosion + writer-starvation livelock — synchronous dispatch, lock released first (-race clean).
- LLMInstance.Available/LastUsed data race (mutex in 1/5 sites) — accessors (-race clean).
- APIKeyStore unsynchronized map → fatal concurrent-map panic (DoS) — RWMutex (-race clean). JWT/rate-limit confirmed already-hardened.

NEW tracked finding (this wave)

- pkg/translator/llm full go test HANGS when api_keys.sh is in-env — a NON-tagged test reaches the network (§11.4.98 violation: unit tests must be deterministic/offline). The wave's new defect tests are httpstest/deterministic; the

hang is in a pre-existing test. FIX: build-tag or httpstest-ize the offending test.
(Bounded `go test -run <name>` works fine.)

Remaining queue (for fresh-session continuation per §11.4.141/ §11.4.127)

- G1: submodule verify-pipeline->server-DB bridge (serial submodule wave; §11.4.119 contends with main builds — conductor-owned, NOT parallel with main-repo subagents).
- G2: batched full ~30-provider verification sweep (one-pass timed out 600s).
- NET: fix the non-tagged network-reaching test in pkg/translator/llm (§11.4.98).
- G3: add OPENAI/ANTHROPIC/etc keys to api_keys.sh (operator) to widen provider coverage.
- More untouched packages to bug-hunt: pkg/storage (deeper), pkg/progress, pkg/report, pkg/logger, cmd/* entrypts, pkg/grpc (the flagged send-on-closed-channel at server.go:403 — events pkg has no Unsubscribe; real follow-up).
- Deferred-broader: markdownToHTML round-trip edges; gemini GenerationConfig hardcoded (same class as fix #2); verification multipass duplicate DB writes.

SHORT resumption sentence

Read docs/CONTINUATION.md then .remember/remember.md, run `git fetch --all`, and **continue** the HelixTranslate "raise-to-enterprise" mandate at **Phase 4 (tests/Challenges/HelixQA to ~100% per type + expand real ebook-translation evidence)** while finishing the Phase-3 tail (Models decision, governance gaps). Binding: anti-bluff §11.4 (real captured evidence, no false PASS), no-force-push §11.4.113, flat snake_case submodules §11.4.28/§11.4.29, every change independently reviewed §11.4.142, containers-first infra §11.4.76, release tags prefixed helix_translate- (§11.4.151). Work subagent-driven (§11.4.70) with ≥ 3 parallel streams (§11.4.103).

Live state anchors (moment-valid)

- **Parent HEAD:** see `git log` (W14b adoption commit via `commit_all.sh`) on main; pushed to BOTH milos85vasic/Translator + HelixDevelopment/HelixTranslate (verified, fast-forward, no force). Session commit chain: b3dc7f9(llm_provider) → e775088 → 8a6af67 → 7331c54 → 76677aa → 18c5137 → de256dd → 9004477 → 13d5e30 → af1ef47 → a9b38fa → 218bfa0 → 1f06bf5 → a0930fc → 3027da2 → b2377af → 6d4ae47.
- **llm_provider HEAD:** b3dc7f9 (W7 CONST-036 propagation block + regenerated html/pdf), pushed to HelixDevelopment/LLMProvider master.
- **Constitution HEAD:** 5e671fe (§11.4.151 added), pushed to all 6 upstreams; parent pointer bumped.
- **Build:** `go build ./... = EXIT 0`. **Total test coverage = 50.7%** (`go tool cover -func total`; see docs/testing/coverage_matrix.md).
- **Submodules (flat, snake_case):** challenges containers constitution doc_processor docs_chain helix_qa llm_orchestrator llm_provider llms_verifier security vision_engine.

- **Release prefix:** HELIX_RELEASE_PREFIX=helix_translate (.env git-ignored, .env.example tracked). First RC: helix_translate-1.0.0-dev-0.0.1 — create ONLY when genuinely green (§11.4.40), operator-gated.
- **Safety backup:** ../helix_translate_gitbackup_20260611T181545Z/repo.git.mirror (pre-rename 311c585).

DONE

Phase 1 — flat snake_case submodules (reviewed GO, pushed): 8 submodules renamed; Go pkg challenges/→pkg/challenge_runner/; docs_chain added; .env release-prefix.go build ./... EXIT 0. **Phase 2 — constitution §11.4.151** release-prefix rule → pushed to all 6 constitution upstreams; parent pointer bumped; inheritance meta-test PASS. **Phase 3 (partial):**

- Security/ plain dir → flat security submodule (proven byte-identical to upstream, zero loss; §11.4.124).
- All submodules + main synced to every upstream (one stale containers gitlab mirror FF'd).
- **docs_chain OPERATIONAL:** .docs_chain/contexts/tracked_docs.yaml registered; real pandoc+weasyprint sync produced docs/CONTINUATION.{html,pdf} + README.{html,pdf}; verify = in-sync. See docs/DOCS_CHAIN.md.
- **Real translation PROVEN (anti-bluff):** EN→Serbian via DeepSeek deepseek-chat, 208 Cyrillic chars, differs from source, no placeholders. Evidence: qa-results/translation/<ts>/ (raw, git-ignored). Asset: test/assets/crow_and_pitcher_en.txt.
- **Fixed a regression I introduced:** challenge scripts' PROJECT_ROOT="\$ {SCRIPT_DIR}/../.." broke when moved 1 level deeper → corrected to ../../..; 3 challenges re-verified PASS (cache_invalidation, model_verification_gate, anti_bluff step1-3).

CLOSED this session (2026-06-13, batch e775088 / llm_provider b3dc7f9 — all pushed, captured evidence)

- **W4**  pkg/security flaky Short-TTL — widened TTL→1min; 50× + -race + CPU-load all PASS.
- **W5**  pkg/distributed 120s timeout — bounded SMTP send (real defect) + in-process test servers; 3× ~48.5s, vet 0.
- **W6**  internal/verifier/discovery 600s timeout — injectable URLs (prod unchanged) + sentinel stub + honest SKIP; 3× ~1s, paired §1.1 mutation proven.
- **W7**  llm_provider CONST-036 propagation block — challenge GREEN; html/pdf regenerated.
- **W8**  anti_bluff_execution_challenge.sh BSD-sed — portable tmpfile sed (§11.4.67); isolated rc=0 proof.
- **W9**  tracked .bak/.backup residue — git-history-investigated (§11.4.124), removed + .gitignore; build 0.
- **W10**  -script latin→Cyrillic — pkg/script was never imported into unified-translator; normalizeScript wired; RED→GREEN 7 tests + 3×.

- **W12**  8 challenge scripts brittle root — `git -C rev-parse --show-toplevel` w/ fallback (§11.4.111); all parse:OK root:OK.
- **D2**  (wave2, 7331c54) test/unit intermittent FAIL — TestVerifier/EventEmission timing race (50ms Sleep vs goroutine-dispatched handler) + data race (proven -race). Mutex eventCollector + bounded poll. BEFORE 1/10 FAIL+race → AFTER -race -count=10 10/10 PASS.
- **W2-A**  (wave2) cmd/unified-translator coverage 3.2%→18.0%, mutation-verified; + latent dangling-pointer fix (Steps []TranslationStep→[]*TranslationStep).
- **W2-B**  (wave2) pkg/storage coverage 32.9%→35.3% (Redis cache-key helpers 0%→100%, real SQLite round-trips), mutation-verified, daemon paths honest-SKIP.
- **wave3** (18c5137): **W3** pkg/translator/llm stress/chaos + **FIXED real prod data race** in BaseTranslator cache+stats (RWMutex, -race ×3); **W2** sshworker 19.6%→39.7%; **W2** batch 77.2%→90.6%.
- **wave4** (de256dd): **sshworker progress-map leak** on ctx.Done() fixed (defer cleanup, RED 300/300→GREEN -race); **D3** pkg/translator{,/llm} test-only EventBus races fixed (asyncFlag); **D7** pkg/events TestEventBus_Subscribe race fixed; **W2** verification 64.0%→73.5%; **W2** script (already 100% stmt) +11 behavioral/digraph tests.
- **D5** (9004477) **FIXED real bug** pkg/verification parseNote: CONTENT-only note (no IMPLICATIONS) silently dropped → finalize Content; RED→GREEN, mutation-verified.
- **D6** (9004477) **FIXED** verified_factory map-order flake → order-independent ElementsMatch; 10/10.
- **wave5** (af1ef47): **W14** scripts/commit_all.sh wrapper (no-force/multi-upstream/FF-only/quiescence/explicit-pathspect/bg-push) + hermetic selftest 13/0 (conductor-run verified) — NOT yet wired to real remotes (line-by-line review pending); **W2** format 0%→97.6%, markdown 73.9%→80.1%, coordination 0%→92.9% + **GENUINE nil-guard fix** in TranslateWithRetry (mutation-verified: revert→nil-panic). §11.4.147: format+coordination were rate-limit-crashed at report stage; conductor verified+adopted their work (not lost, not blindly trusted).

Session total: ~39 items resolved (+ D10 pkg/distributed FULLY -race-clean 9→0 races [4 sources incl. FallBackManager.Stop() lifecycle fix]; + D9 hardware parser testability 44.6→68.4%)

Prior: ~37 items (+ models bridge wiring digital.vasic.models into the system; + D12 REAL CORS auth-bypass vuln fixed; + D11 doc)

Prior: ~34 items (+ W15: real Postgres+Redis on-demand integration proven; PG connection-pool DoS bug found+fixed; containers StartRedis helper added)

Prior: 31 items (+ wave7: grpc 0→53.3% bufconn, api 55→62.2% httpstest, security +14 adversarial tests — all real-protocol/anti-bluff, auth confirmed hardened)

Prior: 29 items (+ wave6: websocket race-fix, hardware/distributed chaos tests, D10#2 api_logger PRODUCT data-race fix)

Earlier session total: 26 items (W4-W10, W12 + D2 + W2×6[unified-translator/storage/batch/sshworker/verification/script] + W3 translator + sshworker-leak + D3 + D7 + D5 + D6). Real prod bugs fixed: translator cache race, SMTP no-deadline, dangling step-ptr, sshworker map-leak, event races (D2/D3/D7), note-content loss (D5).

OPEN workable items (queue — many parallelizable, disjoint scope)

#	Item	Type	Notes / evidence
W1	Models/ (digital.vasic.models) orphan module	RESOLVED- via-bridge	Operator directed "create bridge": pkg/modelsbridge now CONSUMES digital.vasic.models (go.mod require+replace=>./Models), contract-tested 100%, mutation-verified (146b2b5). No longer orphan. W16 (convert to git submodule) still optional/operator-gated. Done: unified-translator 18%, storage 35.3%, batch 90.6%, sshworker 39.7%, verification 73.5%, script(100%+depth), translator/llm(+stress/chaos). Also grpc 53.3% (bufconn), api 62.2% (httptest), security 84.8% +adversarial, hardware/distributed chaos. NOTE: gRPC/api-server/server/redis/postgres at 0% are integration-infra-gated (need real daemons) — W15 containers-first unlocks these; honest deferral per package.
W2	Coverage → ~100% (§11.4.27); ongoing	Task	
W3	Missing test types: chaos, ddos, scaling, ui, ux, full-automation (in-module)	Task	§11.4.85 chaos/stress mandatory.
W11	Wire more docs_chain contexts (Issues/Fixed/Status once they exist)	Task	docs_chain proven operational.
W13	.gitmodules section labels still PascalCase	Task (cosmetic)	Needs per-submodule .git/modules/* gitdir move.
W14	No project commit wrapper scripts/commit_all.sh (§11.4.22)	Task	IN-FLIGHT (wave5): multi-upstream push + no-force §11.4.113 + explicit-paths spec + quiescence + background-push §11.4.88 + hermetic self-test. Review before wiring to real remotes.

#	Item	Type	Notes / evidence
W15	Containers-first infra (§11.4.76)	Feature	storage DONE: real Postgres+Redis on-demand integration via containers/pkg/brokertest (StartPostgres + new StartRedis), build-tagged; pkg/storage CRUD+chaos exercised real DBs; found+fixed PG unbounded-pool DoS (default 25/5). REMAINING: api/grpc DB+LLM <i>pipeline</i> paths need operator-gated LLM keys/SSH (can't autonomously). Pattern + helpers in place for any future slice.
W16	Convert Models/ →models submodule after W1 resolved	Task	Blocked on W1.
D1	docs_chain + security CONSTITUTION.md lack literal CONST-036	Task	§11.4.118 discovery. They use §11.4.X cascade scheme (not CONST-NNN); challenge doesn't check them. Governance-design decision needed — do NOT blindly inject CONST-NNN (§11.4.6). konjugacija→коњугација (should be коњугација); injekcija, надживети similar. Greedy digraph matching; real fix needs morpheme/dictionary model (high blast radius) — operator-aware decision. Pinned with KNOWN-LOSSY tests.
D4	pkg/script Latin→Cyrillic morpheme-boundary mis-transliteration	Bug (open)	Frontmatter now must begin at first non-blank line; later --- = HR/ chapter separator. RED(1ch)→GREEN(2ch), mutation-verified. Minor →dash numbering remains pinned/unfixed (D-list, low priority).
D8	 FIXED (218bfa0) pkg/markdown bare-leading --- data-loss	Bug (fixed)	
D9	 FIXED (fc66c62) hardware parser testability	Task	Extracted 12 pure parsers from exec callers (behavior-preserving);

#	Item	Type	Notes / evidence
			44.6→68.4%, mutation-verified.
D10	3 baseline data races (FACT, captured -race)	Bug	#2 api logger PRODUCT race  FIXED (3027da2, snapshot, mutation-verified). #1 ssh_pool + #3 performance ConnectionPool.  ALL FIXED (fc66c62): ssh_pool options API, perf locked-map + count-snapshot, FallbackManager.Stop() lifecycle. pkg/distributed now 0 races.
D11	 FIXED (b8a24f5) CLAUDE.md mislocated CORS under pkg/security	Task	Corrected: CORS is server-layer (cmd/server corsMiddleware + internal/config). cmd/server corsMiddleware reflected ARBITRARY origin + Allow-Credentials:true under default ["*"] → any site could make
D12	 FIXED (b8a24f5) CORS auth-bypass vuln	Bug(security)	credentialed cross-origin requests. Fixed: wildcard→literal * no-creds; creds only for specific allowlist. RED→GREEN, mutation-verified.
			Independent review: guards sound (force-reject full-argv, FF-only via merge-base --is-ancestor, multi-upstream URL-dedup, explicit-pathspect, quiescence).
W14b	 DONE — reviewed + adopted	Task	FOUND+FIXED cross-platform hardening: mutation grep lacked -I → GNU grep (Linux) false-aborts on binary .pdf/.html pathspects (BSD grep doesn't; §11.4.81). Self-test 13/0. ADOPTED for normal source/doc commits. Dogfood surfaced finding #2: the §11.4.84 residue

#	Item	Type	Notes / evidence
			scan false-positives on files that legitimately CONTAIN marker strings (the script's own MUTATION_MARKERS def, governance docs, marker-referencing tests) — inherent to grep-based detection; §11.4.84 has no escape hatch (correct), so commit such files via manual explicit-path git. commit_all.sh is for the common case.

NEXT phases (priority order)

- **Phase 4** — drive W2–W8 + W10: per-type test suites to ~100% (§11.4.27), real Challenges + HelixQA bank execution with captured evidence, expand real ebook translations (all formats, whole chapters, anti-bluff §11.4/§11.4.69). Subagent-driven, ≥3 parallel streams.
- **Phase 5** — containers-first infra (W15); docs_chain context expansion (W11); full docs (guides/diagrams/SQL/templates) + enterprise Website.
- **Phase 6** — RC tag helix_translate-1.0.0-dev-0.0.1 across all submodules + main ONLY when genuinely green (§11.4.40), operator-confirmed.

Binding constraints

Anti-bluff §11.4 (real captured evidence, no false PASS) · no-force-push §11.4.113 (merge-onto-latest-main) · multi-upstream push §2.1 · flat snake_case §11.4.28/§11.4.29 · every change independently reviewed §11.4.142 · host-safety §12 (no suspend/poweroff) · no silent component removal §11.4.122 · containers-first §11.4.76 · release prefix §11.4.151 · subagent-driven default §11.4.70 · ≥3 parallel streams §11.4.103.

Gotchas

- /Volumes/T7 is **case-sensitive HFS+** (NOT the internal disk — verify on the actual volume, §11.4.6).
- No scripts/commit_all.sh; stage explicit paths, never git add -A; never git add helix_qa (nested third-party pointer drift).
- PIPESTATUS empty in this shell (zsh); use \$? directly.
- Mutation challenges can leave .go.backup/.bak residue — scan + clean before commit (§11.4.84).